United States International University-Africa

EDUCATION TO TAKE YOU PLACES

SCHOOL OF SCIENCE & TECHNOLOGY:
DEPARTMENT OF DATA SCIENCE AND ANALYTICS
SEMESTER: FALL 2024
COURSE: MTH 2215 (B): DISCRETE MATHEMATICS
ASSIGNMENT 2
INSTRUCTER: PROF. (DR.) JONNALAGADDA VENKATESWARA RAO

DATE: **9th** November 2024 **DUE DATE: 16th** November 2024

Marks: 25

Instructions:

After filling the answers, **only one document** should be submitted by the **group via blackboard**. Submission of the document via email & WhatsApp will be rejected. The submission will not be allowed after the deadline, 16-NOVEMBER-2024.

Answers should be written in detail with workouts wherever it is necessary. Marks will be deducted if there is no appropriate workout.

Don't change the document. Use the same document with USIU logo to answer. The document should be submitted as a single PDF file (All pages will be in a single file with name & id number on the first page). JPG (PHOTS) / ZIP files are not allowed & such documents will be rejected.

S.NO	NAME OF THE STUDENT (WRITE THE NAME OF THE GROUP LEADER FIRST)	ID NUMBER
1		
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1. (i) How many different car number plates are possible with 3 letters followed by 3 digits? (0.5 MARK)

(ii) How many of these number plates begin with ABC? (0.5 MARK)

2. (i) In how many ways can 6 people be arranged in a row? (0.5 MARK)

(ii) How many arrangements are possible if only 3 of them are chosen? (0.5 MARK)

3. A mathematics debating team consists of 4 speakers.
(i) In how many ways can all 4 speakers be arranged in a row in a row for a photo?
(ii) How many ways can the captain and vice-captain be chosen? (1 Mark)

4. There are 4 horses in a race. (i) In how many different orders can the horses finish
(ii) How many trifectas (1^{st}, 2^{nd} and 3^{rd}) are possible? (1 Mark)

5. In how many ways can 5 boys and 4 girls be arranged in a bench if (i) there are no restrictions? (ii) boys and girls are alternative? (1 MARK)

6. In how many ways can 5 boys and 4 girls be arranged on a bench if (i) boys and girls are in separate groups? (ii) Anne and Jim wish to sit together? (1 Mark)

7. In how many different arrangements of the word PARRAMATTA are possible? (1 Mark)

8. Find the coefficient of x in the expansion of $(1 - 3x + x^2)(1 - x)^{16}$ (1 Mark)

9. Find the coefficient of x^{15} in the expansion of $(x - x^2)^{10}$ (1 Mark)

10. Find the value of a if the 16^{th} and 17^{th} terms of the expansion $(2 + a)^{50}$ are equal. (1 Mark)